Huawei Access Node HEB-ACC-40: Procedure for Resolving Major Hardware/Environmental Faults (Alarm Code: ALARM_9430)

Node: HEB-ACC-40 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_9430

Description

The failure domain of hardware/environmental faults on the Huawei Access Node HEB-ACC-40, specifically in the context of ASIC memory parity errors, line card crashes, and power supply redundancy loss, arises from the intricate interplay between the node's hardware components and environmental factors. The Advanced Symmetric Cryptography (ASC) module, which is responsible for encrypting and decrypting data packets, is prone to errors due to its complex operation and the high-speed data transfer rates. Furthermore, the Line Card (LC) units, which provide the physical interfaces for data communication, may experience crashes due to overheating, electrical surges, or software bugs. Lastly, the power supply redundancy loss can occur due to a faulty PSU or a mismatch in the power supply configuration. These hardware malfunctions can trigger alarm code ALARM_9430, indicating a major failure in the system.

Pre-requisite Checks

- Ensure that the remote hands team is dispatched to the Hebbal Edge site to perform physical checks on the node.

- Verify the status of the redundant links to ensure that they are functioning correctly.

- Check the environmental conditions, such as temperature and humidity, to ensure that they are within the recommended range.

Potential Root Causes

Cause	Probability
ASIC Memory Parity Errors	High
Line Card Crashes	Medium
Power Supply Redundancy Loss	Low

Diagnostic Commands & Expected Outputs

```
Command: display current-configuration, Expected Output: Current configuration of the node, Description: This command will display the current configuration of the node, including the settings of the ASC module, line cards, and power supplies. By analyzing the output, we can identify any discrepancies or errors that may be contributing to the failure.
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```
Command: debug asc error, Expected Output: ASC error messages, Description: This command will display the error messages related to the ASC module. By analyzing the output, we can identify the source of the errors and take corrective action.
```

Resolution Steps

Step 1: Step: Replace the faulty ASIC module, Command: replace asc-module {slot-number}, Description: To resolve the ASIC memory parity errors, we need to replace the faulty ASIC module. This can be done by using the 'replace asc-module' command, specifying the slot number where the faulty module is located.

Step 2: Step: Replace the faulty Line Card, Command: replace line-card {slot-number}, Description: To resolve the Line Card crashes, we need to replace the faulty Line Card. This can be done by using the 'replace line-card' command, specifying the slot number where the faulty Line Card is located.

Step 3: Step: Replace the faulty Power Supply, Command: replace power-supply {slot-number}, Description: To resolve the Power Supply redundancy loss, we need to replace the faulty Power Supply. This can be done by using the 'replace power-supply' command, specifying the slot number where the faulty Power Supply is located.

Post-Check Verification

```
Command: verify configuration, Expected Output: Verified configuration of the node, Description: This command will verify that the configuration changes made during the resolution process have been successfully applied.

Command: display current-configuration, Expected Output: Current configuration of the node, Description: This command will display the current configuration of the node, including the settings of the ASC module, line cards, and power supplies. By analyzing the output, we can confirm that the resolution steps have been successfully completed.
```

Roll-back Procedures

- Step: Restore the original configuration, Command: undo replace asc-module {slot-number}, Description: If the replacement of the faulty ASIC module does not resolve the issue, we need to restore the original configuration by using the 'undo replace asc-module' command, specifying the slot number where the faulty module was replaced.

- Step: Restore the original configuration, Command: undo replace line-card {slot-number}, Description: If the replacement of the faulty Line Card does not resolve the issue, we need to restore the original configuration by using the 'undo replace line-card' command, specifying the slot number where the faulty Line Card was replaced.

- Step: Restore the original configuration, Command: undo replace power-supply {slot-number}, Description: If the replacement of the faulty Power Supply does not resolve the issue, we need to restore the original configuration by using the 'undo replace power-supply' command, specifying the slot number where the faulty Power Supply was replaced.